Arthi S

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Citizenship: India

Education

Doctor of Philosophy — Oct. 2020 – Ongoing
Electrical Communication Eng., Indian Institute of Science (IISc) Bangalore — *CGPA: 8.7*
Research supervisor: Prof. Neelesh B. Mehta

Master of Technology — 2015 – 2017
Electrical Eng., Indian Institute of Technology (IIT) Madras — *CGPA: 8.7*

Bachelor of Engineering — 2011 – 2015
Electronics & Communication Eng., College of Engineering Guindy — *CGPA: 9.4*

Research Interests

Wireless communications, Next-generation WLANs and 5G/6G systems, Stochastic modeling of communication systems, and Cross-layer performance analysis

Research Experience

PhD Research Details

Focus: Modeling, analysis, and optimization of key features in next-generation IEEE 802.11 (Wi-Fi) systems, with a focus on cross-layer interactions between the physical and MAC layers

1. *Hybrid access OFDMA in IEEE 802.11ax/Wi-Fi 6:*
   - Designed an optimal policy that dynamically allocates resources for random and scheduled access, addressing the fundamental trade-off between saturation throughput and average access delay
   - Developed a cross-layer analysis that accurately characterizes the saturation throughput in time-varying channels to account for small-scale fading and feedback delays
2. *Multi-link operation with uplink OFDMA random access in IEEE 802.11be/Wi-Fi 7:*
   - Identified the operating regimes in which the shortest/longest back-off rule yielded a higher saturation throughput for multi-link operation with OFDMA-based random access
   - Developed a novel Markov model for the non-simultaneous multi-link operation accounting for restricted channel access due to hardware limitations
   - Characterized the throughput and access delay performance for different multi-link operation modes with and without legacy users
   - Determined the optimal contention parameters for the access point that ensured a fair co-existence between multi-link and legacy users

M.Tech Thesis Details

Performance evaluation of FBMC vs. OFDM in the context of NB-IoT (Thesis advisor: Prof. R. David Koilpillai)

- Analyzed three techniques of multi-carrier waveform FBMC - cosine modulated multitone, staggered multitone, and filtered multitone. Explored the reduction in out-of-band spectral emissions gained using FBMC as opposed to traditional waveform OFDM

Publications

Journal Publications

[J1] S. Arthi, N. B. Mehta, and C. Singh, "Hybrid access MAC protocol in Wi-Fi: Analysis and optimal resource allocation policy design," *IEEE Trans. Mob. Comput.*, vol. 25, no. 4, pp. 5742-5757, Apr. 2026.

[J2] S. Arthi and N. B. Mehta, "S. Arthi and N. B. Mehta, "Cross-layer throughput analysis of hybrid-access Wi-Fi in time-Varying channels," *IEEE Wireless Commun. Lett.*, vol. 15, pp. 3676-3680, Jun. 2026.

[J3] S. Arthi and N. B. Mehta, "Multi-link operation modes in Wi-Fi 7 with UORA: Throughput and access delay comparisons," *under review in IEEE Trans. Mobile Comput.*, 2026.

[J4] S. Arthi and N. B. Mehta, "Simultaneous multi-link operation in Wi-Fi 7 with UORA: Throughput and delay implications with legacy users," *to be submitted to IEEE Trans. Commun.*, 2026.

Conference Publications

[C1] S. Arthi and N. B. Mehta, "Saturation throughput analysis of hybrid access MAC protocol in IEEE 802.11ax WLANs," in *Proc. IEEE Glob. Commun. Conf (GLOBECOM)*, Cape Town, South Africa, Dec. 2024.

[C2] S. Arthi and N. B. Mehta, "Saturation throughput analysis of UORA-enabled multi-Link operation in IEEE 802.11be Wi-Fi," in *Proc. IEEE Int. Conf. Commun. (ICC)*, Glasgow, UK, May 2026.

[C3] S. Arthi and N. B. Mehta, "Effectiveness of UORA-Enabled Multi-Link Operation in Wi-Fi 7 in the Presence of Legacy Users," *submitted to IEEE Wireless Commun. Netw. Conf. (WCNC)*, 2027.

[C4] K. Tushar Vikash, S. Arthi, and N. B. Mehta, "Saturation Throughput and Mean Access Delay of Enhanced Multi-Link Single-Radio in Wi-Fi 7," *submitted to IEEE Wireless Commun. Netw. Conf. (WCNC)*, 2027.

Industry and Internship Details

Senior Engineer Jul. 2017 – Sep. 2020
Qualcomm India Private Limited, Hyderabad, India

- Part of the team responsible for RF verification for Qualcomm modems across multiple generations
- Developed a suite of PERL-based tools and tests for automatic internal regression checking of the physical layer and RF parameters
- Proficient working knowledge of RF Software test methodologies of the latest mobile handsets for bring-up and certification and conversant with wireless technologies like LTE, WCDMA, CDMA, GSM, and C-V2X
- Recipient of multiple Qualstar awards for contributions to key projects

Student Trainee Jun. – Jul. 2016
Samsung R&D Institute, Bangalore, India

- Study of wireless technologies (1G - 4G), Doppler and CQI estimators.
- Log analysis with various applications (voice call, data, etc.,) in single SIM mobile

Summer Research Fellow May – Jul. 2014
Raman Research Institute, Bangalore, India

- Research Internship under Prof. Temkar N Ruckmongathan, Raman Research Institute, Bangalore
- Worked on a project to study the application of Ramanujan sums transform for image compression which resulted in better image compression than the discrete-cosine and Hadamard transforms

Awards and Achievements

- Recipient of Qualcomm Innovation Fellowship (QIF) in super-winner category 2025 – 2026
- Recipient of Qualcomm Innovation Fellowship (QIF) 2024 – 2025
- Recipient of Prime Minister's Research Fellowship (PMRF) 2021 – 2024
- Recipient of ORION INSTA award for outstanding contributions and efforts in Qualcomm 2018
- Recipient of Indian Academy of Sciences (IAS) Summer Research Fellowship 2014
- District topper in class XII in Virudhunagar District, Tamil Nadu, with a percentage of 98.2 2011

Coursework at IISc

Random Processes, Digital Communication, Matrix Theory, Detection and Estimation Theory, Wireless Communications, Wireless Networks, Next Generation Wireless System Design, and Stochastic Processes and Queuing Theory

Teaching Responsibilities

Teaching Assistant, IISc Bangalore

- Wireless Communications, *Instructor:* Prof. Neelesh B. Mehta Jan. – Apr. 2025

Course instructor, Institute of Smart Structures and Systems (ISSS)

- Modeling and Analysis of Wi-Fi MAC Protocols Sep. – Oct. 2024
- Wireless Networking Mar. – May 2024
- Matrix Analysis Aug. – Jan. 2024

Course instructor, Dayananda Sagar University, Banaglore

- Information Theory in Computer Science Sep. – Dec. 2022

Teaching Assistant, IIT Madras

- Introduction to Wireless and Cellular Communications, *Instructor:* Prof. R. David Koilpillai Jan. – Apr. 2017

Reviewer Assignments

- IEEE Transactions on Mobile Computing
- IEEE Transactions on Vehicular Technology
- IEEE Wireless Communication Letters